

EXPERIMENT - 15

1. Problem Statement

Write a menu-driven C program using `switch-case` to input a positive integer and perform one of the following operations based on the user's choice:

- **Option 1:** Check whether all digits are even.
E.g. 1: Number: 12345, Output: All digits not even
E.g. 2: Number: 24680, Output: All digits are even
- **Option 2:** Check whether all digits are prime.
E.g. 1: Number: 2357, Output: All digits are prime
E.g. 2: Number: 2379, Output: All digits are not prime

2. Logic Used

Both the options need the digits of the number one by one. For this I used two statements:

Statement	What it does
<code>digit = temp % 10;</code>	Gives the last digit of the number
<code>temp = temp / 10;</code>	Removes the last digit

The while loop keeps running until `temp` becomes 0. So for 12345 the digits come out in the order 5, 4, 3, 2, 1. The order does not matter here because we only have to check each digit, we do not have to keep them in sequence.

For both the options I used a flag variable. In the beginning I assume the answer is "yes" and set the flag to 1. Inside the loop, if any one digit fails the condition, the flag is made 0. After the loop is over, the flag gives the final answer.

Option	Condition checked on each digit
1. All even	If <code>digit % 2 != 0</code> then the digit is odd, so <code>all_even</code> is made 0
2. All prime	If the digit is not 2, 3, 5 or 7 then it is not a prime, so <code>all_prime</code> is made 0

Prime digits

A digit can only be from 0 to 9. Out of these, only 2, 3, 5 and 7 are prime numbers.

Digit	0	1	2	3	4	5	6	7	8	9
Prime?	No	No	Yes	Yes	No	Yes	No	Yes	No	No

So there is no need to write a separate prime checking loop for every digit. A simple comparison of the digit with 2, 3, 5 and 7 is enough. This also makes the program shorter.

1 is not a prime number, because a prime number must have exactly two factors and 1 has only one factor. 0 is also not a prime. This is the reason 2379 gives "All digits are not prime", since the digit 9 is 3×3 . Also, 0 is an even digit, which is why 24680 gives "All digits are even".

3. Algorithm

```

Step 1  : START

Step 2  : Display the menu
          1. Check whether all digits are even
          2. Check whether all digits are prime

Step 3  : Read choice

Step 4  : Read num

Step 5  : If num <= 0 then
          Print "Error: not a positive integer"
          Go to Step 9
        End If

Step 6  : switch (choice)

Step 7  : case 1:                                [ ALL EVEN ]
          Set all_even = 1, temp = num
          While temp > 0 repeat
              digit = temp % 10
              If digit % 2 != 0 then
                  all_even = 0
              End If
              temp = temp / 10
          End While
          If all_even = 1 then
              Print "All digits are even"
          Else
              Print "All digits not even"
          End If
          break

Step 8  : case 2:                                [ ALL PRIME ]
          Set all_prime = 1, temp = num
          While temp > 0 repeat
              digit = temp % 10
              If digit is not 2 and not 3

```

```

        and not 5 and not 7 then
            all_prime = 0
        End If
        temp = temp / 10
    End While
    If all_prime = 1 then
        Print "All digits are prime"
    Else
        Print "All digits are not prime"
    End If
    break

default:
    Print "Invalid choice"
    break

```

Step 9 : STOP

4. Test Case Table

TC #	Choice	Input Number	Expected Output	Actual Output	Result
TC1	1	12345	All digits not even	All digits not even	PASS
TC2	1	24680	All digits are even	All digits are even	PASS
TC3	2	2357	All digits are prime	All digits are prime	PASS
TC4	2	2379	All digits are not prime	All digits are not prime	PASS
TC5	1	8	All digits are even	All digits are even	PASS
TC6	1	2222	All digits are even	All digits are even	PASS
TC7	2	1	All digits are not prime	All digits are not prime	PASS
TC8	2	20	All digits are not prime	All digits are not prime	PASS
TC9	1	-6	Error message	Error: -6 is not a positive integer.	PASS
TC10	2	0	Error message	Error: 0 is not a positive integer.	PASS
TC11	3	123	Invalid choice message	Invalid choice! Enter 1 or 2.	PASS

Reason for taking these test cases

TC #	Reason
TC1	First example given in the question. The number has odd digits in it.
TC2	Second example given in the question. It also has a 0 in it, so it checks that 0 is taken as an even digit.
TC3	Example given in the question. All the four prime digits 2, 3, 5, 7 are used together.
TC4	Example given in the question. Only the digit 9 is not prime and the answer changes because of it.
TC5	A number of only one digit, to see that the program does not fail for small inputs.
TC6	All the digits are same, to check that the loop runs for every digit and not only the first one.
TC7	To check the digit 1. Many students take 1 as a prime, but it is not, so the output must be "not prime".
TC8	To check the digit 0 in the prime option, because 0 is also not a prime.
TC9	A negative number. The question says positive integer, so it should be rejected.
TC10	Zero is also not a positive integer, so it should be rejected.
TC11	A choice other than 1 and 2, to check the <code>default:</code> part of the switch.

5. Source Code

File name: `even_prime.c` (72 lines)

```
/* Menu-driven program using switch-case
   1. Check whether all digits of a number are even
   2. Check whether all digits of a number are prime */

#include <stdio.h>

int main()
{
    int choice, digit;
    long num, temp;
    int all_even, all_prime;

    printf("===== MENU =====\n");
    printf("1. Check whether all digits are even\n");
    printf("2. Check whether all digits are prime\n");
    printf("===== \n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("Error: %ld is not a positive integer.\n", num);
        return 1;
    }

    switch (choice) {

        case 1:
            all_even = 1;
            temp = num;

            while (temp > 0) {
                digit = temp % 10;
                if (digit % 2 != 0)
                    all_even = 0;
                temp = temp / 10;
            }

            printf("\nNumber : %ld\n", num);
            if (all_even == 1)
                printf("Output : All digits are even\n");
            else
                printf("Output : All digits not even\n");
            break;

        case 2:
            all_prime = 1;
```

```
temp = num;

while (temp > 0) {
    digit = temp % 10;
    if (digit != 2 && digit != 3 && digit != 5 && digit != 7)
        all_prime = 0;
    temp = temp / 10;
}

printf("\nNumber : %ld\n", num);
if (all_prime == 1)
    printf("Output : All digits are prime\n");
else
    printf("Output : All digits are not prime\n");
break;

default:
    printf("\nInvalid choice! Enter 1 or 2.\n");
    break;
}

return 0;
}
```

6. Writing the Program in the Terminal

I wrote the program in the `MyAssignment` folder using the `nano` editor:

```
$ cd ~/Desktop/MyAssignment
$ nano even_prime.c
```

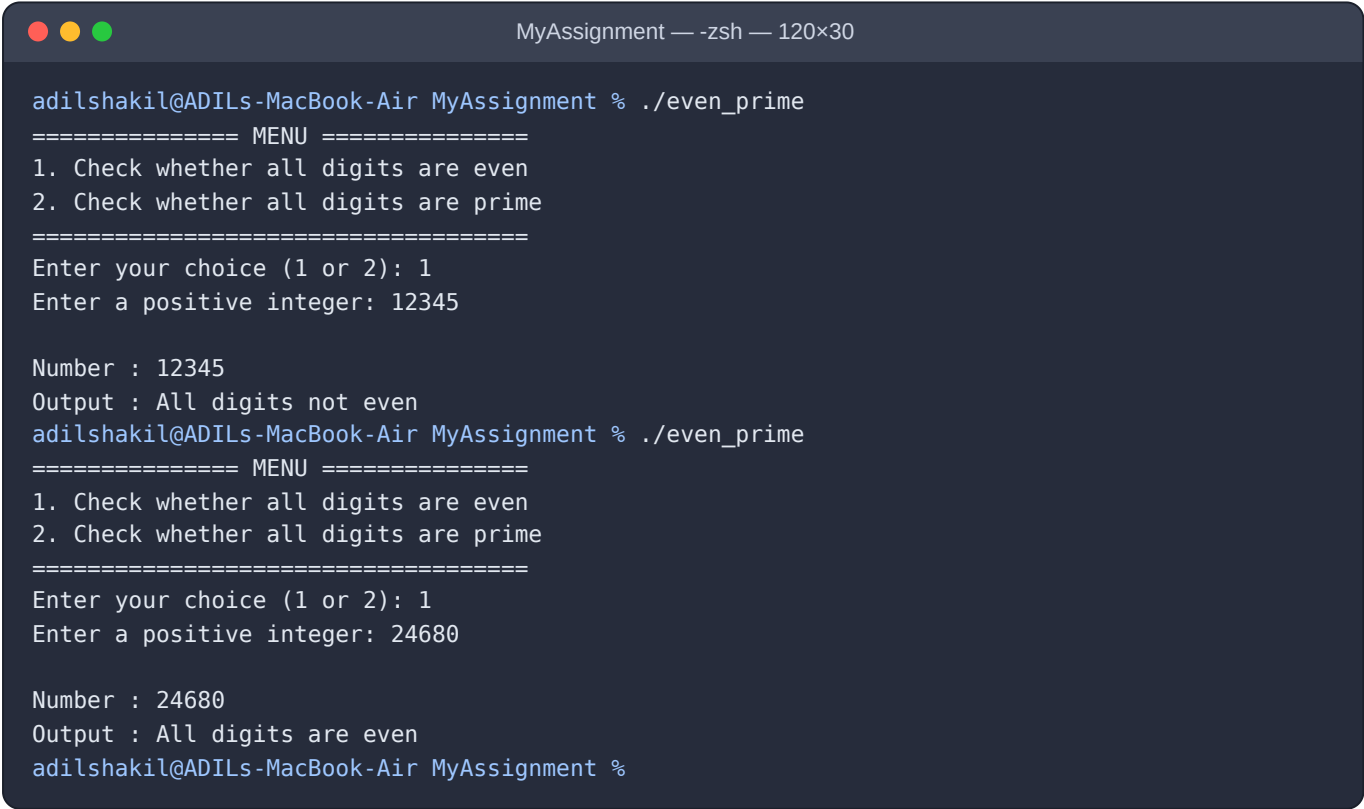
7. Compilation & Execution

Then I compiled it with the GCC compiler and ran it in the Terminal:

```
$ gcc even_prime.c -o even_prime
$ ./even_prime
```

The program compiled in the first attempt itself, without any error or warning.

8. Output

A terminal window titled "MyAssignment — -zsh — 120x30" showing the execution of a C program. The user runs ./even_prime, which displays a menu with two options: "1. Check whether all digits are even" and "2. Check whether all digits are prime". The user selects option 1 and enters the integer 12345. The program outputs "Number : 12345" and "Output : All digits not even". The user then runs ./even_prime again, selects option 1, and enters the integer 24680. The program outputs "Number : 24680" and "Output : All digits are even". The prompt "adilshakil@ADILs-MacBook-Air MyAssignment %" is visible at the end of the terminal output.

```
MyAssignment — -zsh — 120x30

adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 1
Enter a positive integer: 12345

Number : 12345
Output : All digits not even
adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 1
Enter a positive integer: 24680

Number : 24680
Output : All digits are even
adilshakil@ADILs-MacBook-Air MyAssignment %
```

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 2
Enter a positive integer: 2357

Number : 2357
Output : All digits are prime
adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 2
Enter a positive integer: 2379

Number : 2379
Output : All digits are not prime
adilshakil@ADILs-MacBook-Air MyAssignment %
```

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 1
Enter a positive integer: 8

Number : 8
Output : All digits are even
adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 1
Enter a positive integer: 2222

Number : 2222
Output : All digits are even
adilshakil@ADILs-MacBook-Air MyAssignment %
```



```
adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 2
Enter a positive integer: 1

Number : 1
Output : All digits are not prime
adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 2
Enter a positive integer: 20

Number : 20
Output : All digits are not prime
adilshakil@ADILs-MacBook-Air MyAssignment %
```



```
adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 1
Enter a positive integer: -6
Error: -6 is not a positive integer.
adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 2
Enter a positive integer: 0
Error: 0 is not a positive integer.
adilshakil@ADILs-MacBook-Air MyAssignment % ./even_prime
===== MENU =====
1. Check whether all digits are even
2. Check whether all digits are prime
=====
Enter your choice (1 or 2): 3
Enter a positive integer: 123

Invalid choice! Enter 1 or 2.
adilshakil@ADILs-MacBook-Air MyAssignment %
```

9. Conclusion

The program was written, compiled and executed successfully on the Terminal using the GCC compiler. All the 11 test cases gave the same output as expected, so the program is working correctly for both the menu options.

While doing this experiment I found that the two digits which need attention are 0 and 1. The digit 0 has to be counted as an even digit, and the digit 1 is not a prime number. If these two are not handled properly then numbers like 24680 and 1 will give a wrong answer.

The switch-case made the program easy to write because only the case chosen by the user runs, and the `break` statement stops the control from going into the next case.